Inner Product Spaces (note: need lecture before)

Proposition:
Let V be finite dimensional with $\langle \cdot, \cdot \rangle$ an inner product. Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. Let G be a matrix such that $G_{ij} = \langle v_j, v_i \rangle$, then:
$\langle \cdot, \cdot \rangle$ is determined by G. In fact, given $v = a_1 v_1 + \cdots + a_n v_n$ and $b = b_1 v_1 + \cdots + b_n v_n$ in V, then:
(a)

$$\langle v, w \rangle = \begin{bmatrix} \overline{b_1} & \dots & \overline{b_n} \end{bmatrix} G \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

(b) G is Hermitian and positive definite, that is, $G = G^*$, and for all $x \neq \mathbf{0} \in F^{n \times 1}$, then $x^* G x > 0$.
(c) G is invertible.
Proof:
For (a), compute

$$\langle v, w \rangle = \langle a_1 v_1 + \cdots + a_n v_n, b_1 v_1 + \dots b_n v_n \rangle = \sum_{i,j} \langle a_j v_j, b_i v_i \rangle$$

by additivity in both arguments, which is equal to (by conjugate linearity)

$$\sum_{i,j} a_j \overline{b_i} \langle v_j, v_i \rangle = \sum_{i,j} \overline{b_i} G_{ij} a_j = \begin{bmatrix} \overline{b_1} & \dots & \overline{b_n} \end{bmatrix} G \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

finishing the proof of a).
For (b),

$$(G^*)_{ij} = \overline{G_j i} = \overline{\langle v_i, v_j \rangle} = \langle v_j, v_i \rangle = G_{ij}$$

by conjugate symmetry, so $G^* = G$, so G is Hermitian.
Let $x = (a_1, \dots, a_n)^T \in F^{n \times 1}$. So

$$x^* G x = (\overline{a_1}, \dots, \overline{a_n}) G (a_1, \dots, a_n)^T = \langle (a_1, \dots, a_n)^T, (a_1, \dots, a_n)^T \rangle > 0$$

by positivity of the inner product.
(c) is left as an exercise. ■

Definition:
We call the matrix G of the above theorem the matrix of $\langle \cdot, \cdot \rangle$ with respect to the basis $\mathcal{B}$ of V.

Example:
If $\langle \cdot, \cdot \rangle$ is the standard inner product on V with respect to $\mathcal{B}$, then $G = I$.

Conversely,
Proposition:
Let V be a finite dimensional vector space with $\mathcal{B} = (v_1, \dots, v_n)$ a basis. If G is a Hermitian matrix and $x^* G x > 0$ for all $x \neq \mathbf{0} \in F^{n \times 1}$, then there is an inner product $\langle \cdot, \cdot \rangle$ on V such that G is the matrix of $\langle \cdot, \cdot \rangle$ with respect to $\mathcal{B}$.
Proof:
Given $v = a_1 v_1 + \cdots + a_n v_n$, and $w = b_1 v_1 + \cdots + b_n v_n$, define

$$\langle v, w \rangle = \begin{bmatrix} \overline{b_1} & \dots & \overline{b_n} \end{bmatrix} G \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$$

One then checks this is an inner product. ■

Norms and Inner Products

Now take any inner product space $(V, \langle\cdot,\cdot\rangle)$ (not assuming finite dimensional).

Definition:
Let V be an inner product space (not assuming it is finite dimensional). Given $v \in V$, the norm of v is $\|v\| = \sqrt{\langle v, v\rangle}$.

Remark:
One has:
(1) $||v|| \in \mathbb{R}, ||v|| \geq 0$ for all $v \in V$.
(2) If $v \neq \mathbf{0}$, $||v|| > 0$ (since $\langle v, v\rangle$ is positive).
(3) $||\lambda v|| = |\lambda|||v||$ for all $\lambda \in F, v \in V$.
Proof of (3):

$$\langle \lambda v, \lambda v\rangle = \lambda\overline{\lambda}\langle v, v\rangle = |\lambda|^2\langle v, v\rangle \quad \blacksquare$$

Theorem (Cauchy-Schwarz):
Let V be an inner product space. Then for all $v, w \in V$,

$$|\langle v, w\rangle| \leq ||v||||w||$$

Moreover, we have equality if and only if v and w are linearly dependent.
Proof:
This clearly holds if $v = \mathbf{0}$, so assume $v \neq \mathbf{0}$.
Let

$$u = w - \frac{\langle w, v\rangle}{||v||^2}v$$

As an aside, note that u has no component in v's direction:

$$\langle u, v\rangle = \langle w, v\rangle - \frac{\langle w, v\rangle}{||v||^2}\langle v, v\rangle = 0$$

Now we know

$$0 \leq ||u||^2 = \langle w - \frac{\langle w, v\rangle}{||v||^2}v, w - \frac{\langle w, v\rangle}{||v||^2}v\rangle = ||w||^2 - \frac{\overline{\langle w, v\rangle}}{||v||^2}\langle w, v\rangle - \frac{\langle w, v\rangle}{||v||^2}\langle v, w\rangle + \frac{\langle w, v\rangle}{||v||^2}\frac{\overline{\langle w, v\rangle}}{||v||^2}||v||^2$$

$$= ||w||^2 - \frac{\langle w, v\rangle\langle v, w\rangle}{||v||^2} = ||w||^2 - \frac{\overline{\langle v, w\rangle}\langle v, w\ ra}{||v||^2} = ||w||^2 - \frac{|\langle v, w\rangle|^2}{||v||^2}$$

implying

$$||v||^2||w||^2 \geq |\langle v, w\rangle|^2$$

Now suppose $|\langle v, w\rangle| = ||v||||w||$. Then $||u||^2 = 0$ implies $u = \mathbf{0}$ implies $w = \frac{\langle v, w\rangle}{||v||^2}v$, so v and w are linearly dependent.
Conversely, suppose $w = \lambda v$ for some $\lambda \in F$. Then

$$\frac{\langle w, v\rangle}{||v||^2}v = \frac{\langle \lambda v, v\rangle}{v}v = \frac{\lambda||v||^2}{||v||^2}v = \lambda v = w$$

so $w = \dfrac{\langle w, v\rangle}{||v||^2}v$ so by the proof of the main clause, we have equality. $\blacksquare$

Example:
Let V be the vector space of real valued continuous functions on $[0,1]$. Then given $f, g : [0,1] \to \mathbb{R}$, $\langle f, g\rangle = \int_0^1 fgdt \in \mathbb{R}$ is an inner product, with $||f|| = \sqrt{\int_0^1 f^2 dt}$. Then, Cauchy-Schwarz tells us that

$$|\int_0^1 fgdt| \leq \sqrt{\int_0^1 f^2 dt}\sqrt{\int_0^1 g^2 dt}$$

Theorem (Triangle Inequality):
Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Then, $\|v+w\| \leq \|v\| + \|w\|$ for all $v, w \in V$.
Proof:

$$\begin{aligned}||v+w||^2 = \langle v+w, v+w\rangle = ||v||^2 + \langle v,w\rangle + \langle w,v\rangle + ||w||^2 = ||v||^2 + 2\mathrm{Re}(\langle v,w\rangle) + ||w||^2 \\ \leq ||v||^2 + 2|\langle v,w\rangle| + ||w||^2 \leq ||v||^2 + 2||v||||w|| + ||w||^2 = (||v|| + ||w||)^2\end{aligned}$$

where the last inequality is Cauchy-Schwarz. ■

Orthogonality

Definitio n:
Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Then for $v, w \in V$, we say v is orthogonal to w if $\langle v, w\rangle = 0$. If $S \subseteq V$ is a set of vectors, then we say S is orthogonal if $\langle v, w\rangle = 0$ for all $v \neq w \in S$.
If moreover we have $||v|| = 1$ for all $v \in S$, then we say S is orthonormal.

Proposition:
If S is an orthogonal set of nonzero vectors, then S is linearly independent. In fact, for any $v_1, \dots, v_n \in S$ distinct and $w = a_1v_1 + \cdots + a_nv_n$, $a_1, \dots, a_n \in F$, we have $a_j = \frac{\langle w, v_j\rangle}{||v_j||^2}$.
Proof:
The "in fact" is enough, since it shows that anything in $\mathrm{Span}(S)$ can be written in a unique way (the way we wrote down), which is equivalent to linear independence.
Let $w = a_1v_1 + \cdots + a_nv_n$. Therefore,

$$\langle w, v\rangle = \sum_i \langle a_iv_i, v_j\rangle = \sum_i a_i \underbrace{\langle v_i, v_j\rangle}_{0 \text{ if } i \neq j} = a_j\langle v_j, v_j\rangle = a_j||v_j||^2$$

implying $a_j = \dfrac{\langle w, v_j\rangle}{||v_j||^2}$, since $v_j \neq 0$. ■

Corollary:
If $\dim V \leq n$, then the size of any orthogonal set of nonzero vectors is at most n. ■

Theorem (Gram-Schmidt orthogonalization):
Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Let $v_1, \dots, v_n \in V$ be linearly independent vectors. Then there is an orthogonal basis for $\mathrm{Span}(v_1, \dots, v_n)$. (In particular, there is an orthonormal basis, since we can just normalize).
Proof:
We construct $w_1, \dots, w_n$ recursively such that for each $k \leq n$:
(1) $\mathrm{Span}(v_1, \dots, v_k) = \mathrm{Span}(w_1, \dots, w_k)$
(2) $(w_1, \dots, w_k)$ is orthogonal.
At $k = 1$, let $w_1 = v_1$. At $k = m+1$, suppose we have $w_1, \dots, w_m$ satisfying (1) and (2). Define

$$w_{m+1} = v_{m+1} - \sum_{k=1}^{m} \frac{\langle v_{m+1}, w_k\rangle}{||w_k||^2} w_k$$

We then observe:
(i) If $w_{m+1} = 0$, then $v_{m+1} \in \text{Span}(w_1, \dots, w_m) = \text{Span}(v_1, \dots, v_m)$, contradicting linear independence of $v_1, \dots, v_m, v_{m+1}$. Therefore, $w_{m+1} \neq 0$.
(ii) If $1 \leq j \leq m$,

$$\langle w_{m+1}, w_j \rangle = \langle v_{m+1}, w_j \rangle - \langle \sum_{k=1}^{m} \frac{\langle v_{m+1}, w_k \rangle}{||w_k||^2} w_k, w_j \rangle$$

$$= \langle w_{m+1}, w_j \rangle - \sum_{k=1}^{m} \langle \frac{\langle v_{m+1}, w_k}{||w_k||^2} \langle w_k, w_j \rangle$$

By (2), $\langle w_k, w_j \rangle = 0$ unless $k = j$. So, this whole expression is equal to

$$\langle v_{m+1}, w_j \rangle - \frac{\langle v_{m+1}, w_j \rangle}{||w_j||^2} \langle w_j, w_j \rangle = 0$$

Therefore, $(w_1, \dots, w_{m+1})$ is orthogonal.
(iii) We have $w_1, \dots, w_{m+1} \in \text{Span}(w_1, \dots, w_m, v_{m+1})$ (by construction), but by (2) this is $\text{Span}(v_1, \dots, v_m, v_{m+1})$, so by the previous proposition $(w_1, \dots, w_{m+1})$ is a basis for $\text{Span}(v_1, \dots, v_{m+1})$ (it is linearly independent with same dimension using the above, and it spans). Therefore (1) and (2) hold for $(w_1, \dots, w_{m+1})$. ■

Definition:
Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. If $W \subseteq V$ is a finite dimensional subspace with an orthogonal basis $w_1, \dots, w_m$, and $v \in V$, then the vector $w = \sum_{i=1}^{m} \frac{\langle v, w_i \rangle}{||w_i||^2} w_i$ is called the orthogonal projection of v onto W.

Proposition:
The orthogonal projection w of v onto W is the unique vector in W such that $v - w$ is orthogonal to every vector in W. In particular, it does not depend on the choice of orthogonal basis.
Proof:
Proving the first statement, let $w' \in W$. Therefore for the orthogonal basis $w_1, \dots, w_m$ of W, we have $w' = a_1 w_1 + \dots + a_m w_m$ for $a_1, \dots, a_m \in F$. We compute $\langle v - w, w' \rangle$:

$$\langle v - w, w' \rangle = \sum_{i=1}^{m} \overline{a_i} \langle v - w, w_i \rangle$$

$$= \sum_{i=1}^{m} \overline{a_i} \left(\langle v, w_i \rangle - \sum_{j} \frac{\langle v, w_j \rangle}{||w_j||^2} \langle w_j, w_i \rangle \right)$$

$$= \sum_{i=1}^{m} \overline{a_i} (\langle v, w_i \rangle - \langle v, w_i \rangle) = 0$$

where the last line is by orthogonality of $(w_1, \dots, w_m)$.
For uniqueness, suppose $u \in W$ is such that $\langle v - u, w' \rangle = 0$ for all $w' \in W$.
Then, $0 = \langle v - u, w' \rangle = \langle v, w_i \rangle - \langle u, w_i \rangle$. Therefore, $\langle v, w_i \rangle = \langle u, w_i \rangle$ for all $i = 1, \dots, m$. But $u \in W = \text{Span}(w_1, \dots, w_m)$, which is orthogonal. Therefore, by the previous proposition,

$$u = \frac{\langle u, w_1 \rangle}{||w_1||^2} w_1 + \dots + \frac{\langle u, w_m \rangle}{||w_m||^2} w_m$$

$$= \frac{\langle v, w_1 \rangle}{||w_1||^2} w_1 + \dots + \frac{\langle v, w_m \rangle}{||w_m||^2} w_m = w \quad ■$$

Proposition
Given a finite dimensional subspace W of $(V, \langle \cdot, \cdot \rangle)$, define the map $E : V \to W$ by sending $v \in V$ to the orthogonal projection of v onto W. Then E is a projection (called the orthogonal projection onto W).
Proof:
We first check E is linear. Let $\mathcal{B} = (w_1, \ldots, w_m)$ be an orthonormal basis of W, so

$$Ev = \sum_{i=1}^{n} \langle v, w_i \rangle w_i$$

from which it follows

$$E(v_1 + v_2) = \sum_{i=1}^{n} \langle v_1 + v_2, w_i \rangle w_i = \sum_{i=1}^{n} \langle v_1, w_i \rangle w_i + \sum_{i=1}^{n} \langle v_2, w_i \rangle w_i = Ev_1 + Ev_2$$

and similarly $E\lambda v = \lambda Ev$, so E is linear. Next, we show $E^2 = E$. Let $v \in V$. Then note that by definition of the orthogonal complement, $E^2 v$ is the unique vector in W such that $\langle Ev - E^2 v, w \rangle = 0$, since $Ev \in W$, but Ev in place of $E^2 v$ in this expression also satisfies this, so by uniqueness we have $E^2 v = Ev$. ■

Definition
Let $(V, \langle \cdot \rangle)$ be an inner product space, with $S \subseteq V$. The orthogonal complement of S, denoted $S^\perp$, is

$$S^\perp = \{v \in V : \langle v, w \rangle = 0 \ \forall w \in S\}$$

Remark
(1) One sees that $S^\perp$ is a subspace of V for any set S, not just a subspace. Indeed, if $v_1, v_2 \in S^\perp$, then for all $w \in S$,

$$\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle = 0$$

and similarly for λv for $\lambda \in F$.
(2) If $S = V$, then $S^\perp = \{\mathbf{0}\}$, and if $S = \{\mathbf{0}\}$, then $S^\perp = V$.

Proposition
Let $W \subseteq V$ be a finite dimensional subspace. Then for all $v \in V$, $v - Ev \in W^\perp$, with $W^\perp = \mathrm{null}(E)$.
Proof:
The first statement is the definition of Ev. For the second statement, one has $v \in \mathrm{null}(E)$ if and only if $w := Ev = 0$, but w is the unique vector in W such that $v - w \in W^\perp$, so $w = \mathbf{0}$ if and only if $v \in W^\perp$ (since $\mathbf{0} \in W^\perp$, using uniqueness). ■

Corollary
If $W \subseteq V$ is a finite dimensional subspace of V, then $E = \mathrm{range}(E) \oplus \mathrm{null}(E) = W \oplus W^\perp$. ■

Adjoints of Operators

Definition
Let V be a vector space over F. A linear functional on V is a linear map $f : V \to F$, and the space of linear functionals over V is called the dual space of V and is denoted V^*.

Example
If $(V, \langle \cdot, \cdot \rangle)$ is an inner product space, then for any $w \in V$, we have the linear functional given by $v \mapsto \langle v, w \rangle$ for $v \in V$.

Proposition (Riesz Representation Theorem)
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space. Then every linear functional $f \in V^*$ is of the

form $v \mapsto \langle v, w \rangle$ for a unique vector $w \in V$. That is, the map $w \mapsto \langle \cdot, w \rangle$ from V to V^* is a bijection.
Proof:
By Gram-Schmidt, there exists an orthonormal basis $\mathcal{B} = (v_1, \dots, v_n)$ for V. Let $w := \sum_{i=1}^n \overline{f(v_i)} v_i$. We claim $f(v) = \langle v, w \rangle$ for all $v \in V$. Fix $j = 1, \dots, n$. Then

$$\langle v_j, w \rangle = \langle v_j, \sum_{i=1}^n \overline{f(v_i)} v_i \rangle = \sum_{i=1}^n \langle v_j, v_i \rangle = f(v_j)$$

so on the basis $\mathcal{B}$ the linear functionals agree, and thus are equal. Thus, we have surjectivity. On the other hand, for some $w, w' \in V$, suppose f is of the form $\langle \cdot, w \rangle$ and $\langle \cdot, w' \rangle$. Then

$$\langle w - w', w - w' \rangle = \langle w, w \rangle - \langle w, w' \rangle - \langle w', w \rangle + \langle w', w' \rangle = f(w) - f(w) - f(w') + f(w') = 0$$

so $w - w' = 0$, so $w = w'$. ■

Theorem
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space, with $T : V \to V$ a linear operator. There exists a unique linear operator $T^* : V \to V$, called the adjoint of T, such that for all $v, w \in V$, $\langle Tv, w \rangle = \langle v, T^* w \rangle$.
Proof:
Fix $w \in V$ and consider the linear functional $v \mapsto \langle Tv, w \rangle$. By the previous proposition, there exists a unique vector w^* such that $v \mapsto \langle Tv, w \rangle$ is equal to $v \mapsto \langle v, w^* \rangle$. Define $T^* : V \to V$ by $T^* w = w^*$, which is well-defined by uniqueness. First, we check this is linear. Let $\lambda \in F, w_1, w_2 \in V$. Then,

$$\langle v, T^*(\lambda w_1 + w_2) \rangle = \langle Tv, (\lambda w_1 + w_2) \rangle = \overline{\lambda} \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle = \overline{\lambda} \langle v, T^* w_1 \rangle + \langle v, T^* w_2 \rangle = \langle v, \lambda T^* w_1 + T^* w_2 \rangle$$

so the maps $v \mapsto \langle v, T^*(\lambda w_1 + w_2) \rangle$ and $v \mapsto \langle v, \lambda T^* w_1 + \lambda T^* w_2 \rangle$, so by uniqueness in the previous proposition, $T^*(\lambda w_1 + w_2) = \lambda T^* w_1 + T^* w_2$, so T^* is linear. Uniqueness of T^* follows by uniqueness of the previous proposition. ■

Theorem:
Suppose $\mathcal{B} = (v_1, \dots, v_n)$ is an orthonormal basis for V. Then, $[T^*]_{\mathcal{B}} = [T]^*_{\mathcal{B}}$, where the left star is adjoint, the right star is conjugate transpose.
Proof:
Let $A = [T]_{\mathcal{B}}$. We claim $A_{ij} = \langle Tv_j, v_i \rangle$ for all $1 \le i, j \le n$. To see this, use the orthonormality of the basis $\mathcal{B}$ to write

$$Tv_j = \langle Tv_j, v_1 \rangle v_1 + \cdots + \langle Tv_j, v_n \rangle v_n$$

Therefore we see the jth column of A is $(\langle Tv_j, v_1 \rangle, \dots, \langle Tv_j, v_n \rangle)^T$. Therefore, we get that $A_{ij} = \langle Tv_j, v_i \rangle$ as desired. Now let $B = [T^*]_{\mathcal{B}}$. Apply the claim to B to get

$$B_{ij} = \langle T^* v_j, v_i \rangle = \overline{\langle v_i, T^* v_j \rangle} = \overline{\langle Tv_i, v_j \rangle} = \overline{A_j i}$$

for all $1 \le i, j \le n$. Therefore, we see that $B = A^*$. ■

Proposition:
Orthogonal projections are self-adjoint.
Proof:
Let $W \subseteq V$ be a subspace of a finite dimensional inner product space, where E is the orthogonal projection onto W. We want to show $E = E^*$. Fix $v_1, v_2 \in V$ and compute the inner product

$$\langle Ev_1, v_2 \rangle = \langle Ev_1, \underbrace{Ev_2}_{\text{In } W} + \underbrace{(I - E)v_2}_{\text{In } W^\perp} \rangle = \langle Ev_1, Ev_2 \rangle + \langle Ev_1, (I - E)v_2 \rangle = \langle Ev_1, Ev_2 \rangle$$

since $(I - E)v_2 \in W^\perp$. But now doing the same to the other coordinate,

$$\langle Ev_1, Ev_2 \rangle = \langle (I - E)v_1, Ev_2 \rangle + \langle Ev_1, Ev_2 \rangle = \langle v_1, Ev_2 \rangle$$

so E meets the definition of being self-adjoint. ■

Proposition:
Let T, U be linear operators on a finite dimensional inner product space $(V, \langle\cdot,\cdot\rangle)$. Then,
(a) $(T+U)^* = T^* + U^*$
(b) $(\lambda T)^* = \overline{\lambda}T^*$
(c) $(TU)^* = U^*T^*$
(d) $(T^*)^* = T$
Proof:
Make an orthonormal basis for V with Gram-Schmidt and use the rules of conjugate transposes, which determines the adjoints by the previous theorem. ■

Corollary:
Let $T : V \to V$ be a linear operator on a finite dimensional inner product space $(V, \langle\cdot,\cdot\rangle)$ over $\mathbb{C}$.
Let $U_1 = \frac{1}{2}(T + T^*)$, $U_2 = \frac{1}{2i}(T - T^*)$. Then $T = U_1 + iU_2$, $U_1^* = U_1$, $U_2^* = U_2$ (so we can think of U_1 as the "real part" of T, U_2 as the "imaginary part" of T, and $*$ as complex conjugation). ■

Isomorphisms of Inner Product Spaces

Definition:
Let $(V, \langle\cdot,\cdot\rangle_V)$ and $(W, \langle\cdot,\cdot\rangle_W)$ be inner product spaces, and $T : V \to W$ be a linear map. We say that T preserves inner products if for all $v_1, v_2 \in V$, then

$$\langle Tv_1, Tv_2\rangle_W = \langle v_1, v_2\rangle_V$$

If moreover, T is an isomorphism of vector spaces (a linear map that is injective and surjective), then we say that T is an isomorphism of inner product spaces. We write $T : (V, \langle\cdot,\cdot\rangle_V) \to (W, \langle\cdot,\cdot\rangle_W)$ to mean that T is a linear map that preserves the inner product.

Lemma:
Suppose $T : (V, \langle\cdot,\cdot\rangle_V) \to (W, \langle\cdot,\cdot\rangle_W)$ is surjective. Then T is an isomorphism of inner product spaces.
Proof:
We need to show $\ker T = \{0\}$. Let $v \in V$. We know $v = 0$ if and only if $\langle v, v\rangle_V = 0$. But then $\langle v, v\rangle_V = \langle Tv, Tv\rangle_W = 0$ implies $v = 0$ so T is injective, and thus an isomorphism. ■

Remark:
If T preserves inner products, then T preserves the norm, since $||Tv||^2 = \langle Tv, Tv\rangle = \langle v, v\rangle = ||v||^2$, so $||Tv|| = ||v||$.

The converse also holds.

Proposition:
Let $(V, \langle\cdot,\cdot\rangle)$ and $(W, \langle\cdot,\cdot\rangle)$ be inner product spaces. Let $T : V \to W$ be a linear map. Then T preserves inner products if and only if $||Tv|| = ||v||$ for all $v \in V$.
Proof:
We just proved the forward direction.
Conversely, use the polarization identities:
Over $F = \mathbb{R}$, this is

$$\langle v, w\rangle = \frac{1}{4}(||v+w||^2 - ||v-w||^2)$$

Over $F = \mathbb{C}$, this is

$$\langle v, w\rangle = \frac{1}{4}(||v+w||^2 - ||v-w||^2 + ||v+iw||^2 - ||v-iw||^2)$$

Now, suppose T preserves norm, and use these identities. For the case where $F = \mathbb{R}$,

$$\langle Tv, Tw\rangle = \frac{1}{4}(||Tv+Tw||^2 - ||Tv-Tw||^2) = \frac{1}{4}(||T(v+w)||^2 - ||T(v-w)||^2) = \frac{1}{4}(||v+w||^2 - ||v-w||^2) = \langle v, w\rangle$$

and the case for $\mathbb{C}$ is similar. ■

Remark:
If T is an isomorphism of inner product spaces, its inverse also preserves inner products. To see this, write

$$\langle w_1, w_2\rangle = \langle TT^{-1}w_1, TT^{-1}w_2\rangle = \langle T^{-1}w_1, T^{-1}w_2\rangle$$

since T preserves inner products.

Theorem:
Let $(V, \langle\cdot,\cdot\rangle)$ and $(W, \langle\cdot,\cdot\rangle)$ be two inner product spaces of the same finite dimension. Suppose $T : V \to W$ is a linear transformation. Then, the following are equivalent:
(i) T preserves inner products.
(ii) T is an isomorphism of inner product spaces.
(iii) For any orthonormal basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V, then $(Tv_1, \ldots, Tv_n)$ is an orthonormal basis for W.
(iv) Same as (iii), except that it holds if we have that for some orthonormal basis (as opposed to any).
Proof:
(i) implies (ii): We have seen above that if T preserves inner products then it is injective, this follows since $\ker T = \{0\}$, since $||Tv|| = ||v||$. But since $\dim V = \dim W$, an injective linear map on same-dimensional spaces is also surjective, and thus an isomorphism. Therefore it is an isomorphism of inner product spaces (since we know by assumption that T preserves inner products).
(ii) implies (iii): Compute $\langle Tv_i, Tv_j\rangle$: since T preserves inner products,

$$\langle Tv_i, Tv_j\rangle = \langle v_i, v_j\rangle = \delta_{ij}$$

Therefore, $(Tv_1, \ldots, Tv_n)$ is an orthonormal set of nonzero vectors (nonzero since the original basis doesn't have zero vectors, and T is injective), and we know it is a basis of W since T is an isomorphism of vector spaces and thus preserves bases. (Alternatively, since it is a linearly independent set of n vectors in W, it is a basis (since $\dim V = n = \dim W$)).
(iii) implies (iv): Immediate.
(iv) implies (i): Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be an orthonormal basis of V such that $\{Tv_1, \ldots, Tv_n\}$ is an orthonormal basis of W. But now let $v, v' \in V$. Therefore,

$$v = \langle v, v_1\rangle v_1 + \cdots + \langle v, v_n\rangle v_n$$

and

$$v' = \langle v', v_1\rangle v_1 + \cdots + \langle v', v_n\rangle v_n$$

But now compute the inner product:

$$\langle Tv, Tv'\rangle = \langle \sum_{i=1}^{n} \langle v, v_i\rangle Tv_i, \sum_{i=1}^{n} \langle v, v_i\rangle Tv_i\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} \langle v, v_i\rangle \overline{\langle v', v_j\rangle} \langle Tv_i, Tv_j\rangle$$

But $\langle Tv_i, Tv_j\rangle = \delta_{ij}$ since they form orthonormal basis, so this sum is just $\sum_{i=1}^{n} \langle v, v_i\rangle\overline{\langle v', v_i\rangle}$. But now reversing this step with the basis of V that is also orthonormal, the same Kronecker delta holds:

$$\sum_{i=1}^{n}\sum_{j=1}^{n} \langle v, v_i\rangle \overline{\langle v', v_i\rangle} \langle v_i, v_j = \langle v, v'\rangle$$

Therefore we see T preserves inner products. ■

Corollary:
Any two finite dimensional inner product spaces of the same dimension are isomorphic as inner product spaces.
Proof:
Let $(V, \langle \cdot, \cdot \rangle)$ and $(W, \langle \cdot, \cdot \rangle)$ be inner product spaces of dimension n. Use Gram-Schmidt to get orthonormal bases $(v_1, \ldots, v_n)$ and $(w_1, \ldots, w_n)$ for them. Now just take the map from V to W given by $a_1 v_n + \cdots + a_n v_n \mapsto a_1 w_1 + \cdots + a_n w_n$. This is a linear isomorphism (from linear algebra 1), and takes an orthonormal basis of V to an orthonormal basis of W (the two bases we listed), so by the previous theorem this is an isomorphism of inner product spaces. ■

Example:
Take $F = \mathbb{R}$. We have $\mathbb{R}^3$ with the dot product as the standard inner product space.
Take $M_3(\mathbb{R})$, with $\langle A, B \rangle_F = \mathrm{tr}(AB^T)$, the Frobenius inner product. Let W be the subspace of $M_3(\mathbb{R})$ of skew-symmetric matrices, i.e. $A = -A^T$. So W with the Frobenius inner product $\langle \cdot, \cdot \rangle_F$ is an inner product space. It is three dimensional, since all skew symmetric matrices are of the form $\begin{bmatrix} 0 & * & *' \\ -* & 0 & *'' \\ -*' & -*'' & 0 \end{bmatrix}$.
Therefore, by the corollary, $(\mathbb{R}^3, \cdot) \cong (W, \langle \cdot, \cdot \rangle_F)$.
But what is the isomorphism? Try the map $T : \mathbb{R}^3 \to W$ given by $T(x_1, x_2, x_3) = \begin{bmatrix} 0 & x_1 & x_2 \\ -x_1 & 0 & x_3 \\ -x_2 & -x_3 & 0 \end{bmatrix}$.
This doesn't work (quite), but taking $T' : \mathbb{R}^3 \xrightarrow{\cong} W$ given by $\frac{1}{\sqrt{2}} T$ does work.
To verify it preserves inner products, let's check

$$\langle T(x_1, x_2, x_3), T(y_1, y_2, y_2 \rangle_F = \langle \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & x_1 & x_2 \\ -x_1 & 0 & x_3 \\ -x_2 & -x_3 & 0 \end{bmatrix}, \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & y_1 & x_2 \\ -y_1 & 0 & y_3 \\ -y_2 & -y_3 & 0 \end{bmatrix} \rangle_F$$

$$= \frac{1}{2} \mathrm{tr}(\begin{bmatrix} x_1 y_1 + x_2 y_2 & * & * \\ & x_1 y_1 + x_3 y_3 & * \\ * & * & x_2 y_2 + x_3 y_3 \end{bmatrix}) = x_1 y_1 + x_2 y_2 + x_3 y_3 = x \cdot y$$

Definition:
Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space, and $T : V \to V$ is a linear operator. If T is invertible and T preserves the inner product, then we say that T is unitary.
So T being unitary means that $T : (V, \langle , \rangle) \to (V, \langle , \rangle)$ is an automorphism of inner product spaces.

Remark:
If V is finite dimensional, then the following are equivalent characterizations of a linear operator $T : V \to V$ being unitary:
(a) T preserves inner products, since we proved prior that linear maps between vector spaces of the same finite dimension that preserve inner products are isomorphisms.
(b) T preserves norms.
(c) For any orthonormal basis $\mathcal{B}$ of V, $T(\mathcal{B})$ is an orthonormal basis of V
(d) For some orthonormal basis $\mathcal{B}$ of V, $T(\mathcal{B})$ is an orthonormal basis of V.

Theorem:
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space, and let $T : V \to V$ be a linear operator. Then T is unitary if and only if T is invertible and $T^* = T^{-1}$.
Proof:
Suppose T is unitary. Then, by definition, we know it is invertible, and preserves inner products. Let

$v, w \in V$ be arbitrary. Then,

$$\langle Tv, w\rangle = \langle Tv, TT^{-1}w\rangle = \langle v, T^{-1}w\rangle$$

since T preserves the inner product. But T^* is the unique linear operator satisfying $\langle Tv, w\rangle = \langle v, T^*w\rangle$, so $T^{-1} = T^*$.
Conversely, suppose T is invertible with $T^{-1} = T^*$. Let $v, w \in V$. Then,

$$\langle Tv, Tw\rangle = \langle v, T^*Tw\rangle = \langle v, w\rangle$$

where the first equality follows by definition of T^*. Therefore, T is unitary since it preserves inner products. ■

Definition:
A matrix $A \in M_n(F)$ is unitary if it is invertible and $A^{-1} = A^*$ (where this $*$ is the conjugate transpose).

Proposition:
Let $(V, \langle\cdot,\cdot\rangle)$ be a finite dimensional inner product space, and $T : V \to V$ a linear operator. Then T is unitary if and only if for some (equivalently for all) orthonormal basis $\mathcal{B}$ of V, $[T]_\mathcal{B}$ is unitary.
Proof:
Suppose T is unitary, and $\mathcal{B}$ is an orthonormal basis of V. Then, $[T]_\mathcal{B}^* = [T^*]_\mathcal{B}$. But T is unitary, so by the previous theorem, $T^* = T^{-1}$, so we get that

$$[T^*]_\mathcal{B} = [T^{-1}]_\mathcal{B} = [T]_\mathcal{B}^{-1}$$

Therefore, $[T]_\mathcal{B}$ is unitary. Conversely, suppose $\mathcal{B}$ is an orthonormal basis of V such that $[T]_\mathcal{B}$ is unitary. But then

$$[T^*]_\mathcal{B} = [T]_\mathcal{B}^* = [T]_\mathcal{B}^{-1} = [T^{-1}]_\mathcal{B}$$

but this means T^* and T^{-1} have matrices that agree on the same basis, so they are the same linear operator. ■

Lemma:
If $A \in M_n(F)$ is unitary, then $|\det(A)| = 1$. In particular, if $F = \mathbb{R}$, then $\det(A) = \pm 1$.
Proof:

$$1 = \det(I) = \det(A^*A) = \det(A^*)\det(A) = \overline{\det(A)}\det(A) = |\det(A)|^2 \quad ■$$

Examples:
(a) Given a 1×1 matrix $A = [c]$, A is unitary if and only if $|c| = 1$.
(b) Let $A \in M_2(\mathbb{R})$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $A^* = A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, and $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$, so since $ad - bc = \pm 1$, we have that A is unitary if and only if $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ or $A = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$ and $a^2 + b^2 = 1$. Thus, we can write such a matrix in the form $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, which represents the rotation of $\mathbb{R}^2$ by angle θ with respect to the standard basis/inner product.

Proposition:
Let $A \in M_n(F)$. Then A is unitary if and only if its rows are orthonormal with respect to the standard basis/inner product of F^n, and equivalently if and only if its columns are orthonormal with respect to the standard basis/inner product of F^n.
Proof:
First, note that all three statements fail if A is not invertible, since if A is not invertible then its

rows/columns are linearly dependent, but orthonormality implies linear independence (and unitary requires invertibility by definition), so we may assume A is invertible.
We have
$$A \text{ unitary} \iff AA^* = I \iff (i\text{th row of } A) \cdot \overline{(j\text{th row of } A)} = \delta_{ij}$$
where the dot product is the definition of matrix multiplication, and δ_{ij} is the Kronecker delta, corresponding to the entries of the identity. But the dot product of these rows and columns is precisely the standard inner product on F^n, so A is unitary if and only if
$$\langle i\text{th row of } A, j\text{th row of } A \rangle = \delta_{ij}$$
which is the definition of the rows being orthonormal. The proof for columns works identically, or with the proof for the rows combined with the fact that
$$AA^* = I \iff A^*A = I \iff A \text{ unitary} \quad \blacksquare$$

Corollary:
Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space. Suppose $T : V \to V$ is a linear operator. Suppose $\mathcal{B}$ and $\mathcal{B}'$ are orthonormal bases for V. Then the matrices of T with respect to these bases are similar, with $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$, where P is a unitary matrix. We say that $[T]_{\mathcal{B}'}$ and $[T]_{\mathcal{B}}$ are unitarilly similar.
Proof:
Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (w_1, \dots, w_n)$. Let $S : V \to V$ be the linear operator determined by $v_i \mapsto w_i$ for $i = 1, \dots, n$, which is an isomorphism by linear algebra 1. Let $P = [S]_{\mathcal{B}}$. So $[T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}}P$. But we see S is unitary since it takes an orthonormal basis $\mathcal{B}$ to an orthonormal basis $\mathcal{B}'$, so $P = [S]_{\mathcal{B}}$ is unitary, since $\mathcal{B}$ is orthonormal. $\blacksquare$

Orthonormal Diagonalization

Question:
Given an inner product $\langle , \rangle$ on a finite dimensional vector space V, which linear operators are diagonalizable with an orthonormal basis? Equivalently, when does a linear operator have an orthonormal basis of eigenvectors?
Let $(V, \langle , \rangle)$ be a finite dimensional inner product space with $T : V \to V$ a linear operator. When is T orthonormally diagonalizable? i.e. when is there an orthonormal basis of eigenvectors? First, note that diagonalizable does not imply orthonormally diagonalizable.
To see this, suppose T is diagonalizable. Then $V = W_1 \oplus \cdots \oplus W_k$ for W_i's the eigenspaces. Each W_i with the induced inner product has an orthonormal basis by Gram-Schmidt, say $\mathcal{B}_i$, where all vectors in $\mathcal{B}_i$ will be eigenvectors. But then $\mathcal{B} = (\mathcal{B}_1, \dots, \mathcal{B}_k)$ is a basis for V. But if $v \in \mathcal{B}_i$, $w \in \mathcal{B}_j$, with $i \neq j$, then maybe $\langle v, w \rangle \neq 0$. Applying Gram-Schmidt to $\mathcal{B}$ overall doesn't help, since this doesn't preserve the property of vectors being eigenvectors.

Remark:
Suppose T is orthonormally diagonalizable. Let $\mathcal{B} = (v_1, \dots, v_n)$ be an orthonormal basis for V made up of eigenvectors. Then, this is the diagonalization of T, so $[T]_{\mathcal{B}}$ is diagonal with diagonal entries/eigenvalues $\lambda_1, \dots, \lambda_n$. But now notice if we take the adjoint of T, $[T^*]_{\mathcal{B}} = [T]^*_{\mathcal{B}}$, which is a diagonal matrix with diagonal entries/eigenvalues $\overline{\lambda_1}, \dots, \overline{\lambda_n}$. This implies that if $F = \mathbb{R}$, then any orthogonally diagonalizable operator must satisfy $T = T^*$. If $F = \mathbb{C}$, then we must have $TT^* = T^*T$, since diagonal matrices commute with each other.

Definition:
A linear operator T is normal if $TT^* = T^*T$.

Thus, we see that orthonormally diagonalizable operators are normal. Our goal is to prove the converses of these statements.

Examples:
Here are two easy examples of normal linear operators.
(1) Self-adjoint: every linear operator commutes with itself.
(2) Unitary: Every invertible linear operator commutes with its inverse.
There are normal linear operators that are neither self-adjoint, nor unitary, for example take a unitary matrix and multiples of its adjoint, or any 1×1 matrix like $[1+i]$ with adjoint $[1-i]$ (these aren't unitary since their determinant doesn't have absolute value 1).

Remark:
Unlike self-adjointness and unitarity, normality is not preserved under sum/product:

$$(T+S)^*(T+S) = (T^*+S^*)(T+S) = T^*T + T^*S + S^*T + S^*S$$

and we see the cross terms don't commute in general. Similarly

$$(TS)^*(TS) = S^*T^*TS$$

and we see we can't switch the S's.

Proposition:
Suppose T is self-adjoint.
(a) The eigenvalues of T are real.
(b) Eigenvectors corresponding to distinct eigenvalues are orthogonal.
Proof:
(a) Suppose $\lambda \in F$ is an eigenvalue of T. So there is $v \neq 0$ such that $Tv = \lambda v$. Taking

$$\lambda\langle v, v\rangle = \langle \lambda v, v\rangle = \langle Tv, v\rangle = \langle v, T^*v\rangle$$

But T is self-adjoint, so this equals

$$\langle v, Tv\rangle = \langle v, \lambda v\rangle = \overline{\lambda}\langle v, v\rangle$$

implying $\lambda = \overline{\lambda}$ since $\langle v, v\rangle \neq 0$ since $v \neq 0$. Therefore $\lambda \in \mathbb{R}$.
(b) Suppose $\lambda \neq \gamma$, $v \neq 0$, $w \neq 0$, with $Tv = \lambda v$ and $Tw = \gamma w$. Then,

$$\lambda\langle v, w\rangle = \langle \lambda v, w\rangle = \langle Tv, w\rangle = \langle v, T^*w\rangle = \langle v, Tw\rangle = \langle v, \gamma w\rangle = \overline{\gamma}\langle v, w\rangle$$

But by (a), $\overline{\gamma} = \gamma$ (since γ is real), so we see that

$$(\lambda - \gamma)\langle v, w\rangle = 0$$

but λ and γ are distinct, so $\langle v, w\rangle = 0$, so they are orthogonal. ■

Corollary:
If T is self-adjoint and diagonalizable, it is orthonormally diagonalizable. ■

Proposition:
Suppose T is self-adjoint. Then T has an eigenvalue.
Proof:
This follows easily in the complex case from the Fundamental Theorem of Algebra, so assume $F = \mathbb{R}$. Let $\mathcal{B}$ be an orthonormal basis for V, and let A be the matrix of T with respect to $\mathcal{B}$. Consider $\text{char}(T) \in \mathbb{R}[x] \subseteq \mathbb{C}[x]$. Therefore, by the Fundamental Theorem of Algebra, $\text{char}(T)$ has a root $c \in \mathbb{C}$.

Consider $\widetilde{T} : \mathbb{C}^n \to \mathbb{C}^n$, where $n = \dim V$ such that $[\widetilde{T}] = A$. Thus, we see that $\mathrm{char}(\widetilde{T}) = \det(xI - A) = \mathrm{char}(T)$.
Then, working with the standard inner product on $\mathbb{C}^n$, we have $[\widetilde{T}^*] = [\widetilde{T}]^*$ since the standard basis is orthonormal, which equals $A^* = [T]^*_{\mathcal{B}} = [T^*]_{\mathcal{B}}$ since $\mathcal{B}$ is orthonormal, but this equals $[T]_{\mathcal{B}} = A = [\widetilde{T}]$ since T is self-adjoint, so we see $\widetilde{T}$ is self-adjoint.
Thus, by the previous proposition, all of $\widetilde{T}$'s eigenvalues are real, so $c \in \mathbb{R}$ since c is a root of the characteristic polynomial of $\widetilde{T}$ and thus an eigenvalue. Therefore, since c is a real root of the characteristic polynomial of T, it must be a (real) eigenvalue, so T has an eigenvalue (and in fact this argument shows T has n real eigenvalues). ■

Lemma:
If $W \subseteq V$ is T-invariant, then $W^\perp$ is T^*-invariant.
Proof:
Fix $v \in W^\perp$. Let $w \in W$ be arbitrary.

$$\langle T^* v, w\rangle = \overline{\langle w, T^* v\rangle} = \overline{\langle Tw, v\rangle} = 0$$

since $Tw \in W$ since $w \in W$, W is T-invariant, and $v \in W^\perp$. ■

This lemma and the previous theorem gives us our main result:

Theorem (Spectral Theorem):
If T is self-adjoint, then there exists an orthonormal basis of eigenvectors of T.
Proof:
By induction on $\dim V$.
By the previous theorem, T has an eigenvector, v. Let $v_1 = \frac{v}{||v||}$ be its normalization. Let $W = \mathrm{Span}(v_1)$. Note that v_1 is also an eigenvector for T. If $W = V$ we are done.
Thus, suppose $W \subsetneq V$. By the lemma, we know $W^\perp$ will be T^*-invariant, since W is T-invariant (since v_1 is an eigenvector), so since T is self-adjoint, we have that $W^\perp$ is T-invariant.
Therefore, $V = W \oplus W^\perp$ is a T-invariant direct sum decomposition.
We claim T's restriction to $W^\perp$ is still self-adjoint. Indeed, this follows because W and $W^\perp$ being T-invariant allows the restrictions of T and T^* to be well-defined: fix $v_1, v_2 \in W^\perp$. Then,

$$\langle v_1, Tv_2\rangle = \langle v_1, T^* v_2\rangle = \langle Tv_1, v_2\rangle$$

so $(T|_{W^\perp})^* = T|_{W^\perp}$, proving the claim.
Therefore, $\dim W^\perp = \dim V - 1$, so by the induction hypothesis, $W^\perp$ has an orthonormal basis $(v_2, \ldots, v_n)$ of eigenvectors for $T|_{W^\perp}$. Therefore, by properties of the direct sum, $\mathcal{B} = (v_1, v_2, \ldots, v_n)$ is a basis for $V = W \oplus W^\perp$ and $v_1, \ldots, v_n$ are eigenvectors. ■

Corollary:
If $F = \mathbb{R}$, then T is self-adjoint if and only if T has an orthonormal basis of eigenvectors.

Recall:
If $F = \mathbb{R}$, and $\mathcal{B}$ is an orthonormal basis, then T is self-adjoint if and only if the matrix of T with respect to $\mathcal{B}$ is symmetric. Therefore, we can check if an operator is self-adjoint by checking its matrix in an orthonormal basis and verifying if it is symmetric.

Now we shift focus and consider normality, i.e. $TT^* = T^*T$.

Lemma:
Let $T : V \to V$ be a normal linear operator. Then,
(1) For any $v \in V$, $||Tv|| = ||T^*v||$.

(2) Any polynomial in T is also normal.
Proof:
To prove (1), note

$$||Tv||^2 = \langle Tv, Tv\rangle = \langle v, T^*Tv\rangle = \langle v, TT^*v\rangle = \langle v, T^{*^*}T^*v\rangle = \langle T^*v, T^*v\rangle = ||T^*v||^2$$

To prove (2), let $p = a_0 + a_1x + \cdots + a_nx^n \in F[x]$ be a polynomial. Then,

$$p(T)^* = (a_oI+\cdots+a_nT^n)^* = \overline{a_o}I+\overline{a_1}T^*+\overline{a_2}(T^2)^*+\cdots+\overline{a_n}(T^n)^* = \overline{a_0}I+\overline{a_1}T^*+\overline{a_2}(T^*)^2+\cdots+\overline{a_n}(T^*)^n = \overline{p}(T^*)$$

where $\overline{p}$ is obtained from p by taking the complex conjugate of its coefficients. Therefore, $p(T)^*$ is a polynomial in T^*. But since T and T^* commute, any polynomials in them commute, specifically $p(T)$ and $p(T)^*$. ■

Definition:
A matrix $A \in M_n(F)$ is normal if $AA^* = A^*A$.

Lemma:
If $\mathcal{B}$ is an orthonormal basis for T, then T is normal if and only if $[T]_\mathcal{B}$ is normal.
Proof:
Since $\mathcal{B}$ is orthonormal, $[T^*]_\mathcal{B} = [T]_\mathcal{B}^*$. Thus,

$$\begin{aligned} T \text{ normal} &\iff T^*T = TT^* \\ &\iff [T^*T]_\mathcal{B} = [TT^*]_\mathcal{B} \\ &\iff [T^*]_\mathcal{B}[T]_\mathcal{B} = [T]_\mathcal{B}[T^*]_\mathcal{B} \\ &\iff [T]_\mathcal{B}^*[T]_\mathcal{B} = [T]_\mathcal{B}[T]_\mathcal{B}^* \quad \text{Since } \mathcal{B} \text{ orthonormal} \\ &\iff [T]_\mathcal{B} \text{ normal} \quad ■ \end{aligned}$$

Proposition:
Suppose T is normal. Then $v \in V$ is an eigenvector of T with eigenvalue λ if and only if $\overline{\lambda}$ is an eigenvalue for T^* and v is an eigenvector corresponding to $\overline{\lambda}$ for T^*.
Proof:

$$\begin{aligned} Tv = \lambda v &\iff (T-\lambda I)v = 0 \\ &\iff ||(T-\lambda I)v|| = 0 \\ &\iff ||(T-\lambda I)^*v|| = 0 \quad \text{By 2 lemmas previous (since } T-\lambda I) \text{ is a polynomial in } T) \\ &\iff ||(T^*-\overline{\lambda}I)v|| = 0 \\ &\iff (T^*-\overline{\lambda})v = 0 \\ &\iff T^*v = \overline{\lambda}v \quad ■ \end{aligned}$$

Proposition:
Suppose $\mathcal{B}$ is an orthonormal basis of V such that $[T]_\mathcal{B}$ is upper triangular. Then T is normal if and only if $[T]_\mathcal{B}$ is diagonal.
Proof:
Recall that T is normal if and only if $[T]_\mathcal{B}$ is normal (since $\mathcal{B}$ is orthonormal). This proves the reverse direction, since a diagonal matrix will always commute any other diagonal matrices, such as its conjugate transpose which is also diagonal.
In the forward direction, let $\mathcal{B} = (v_1, \ldots, v_n)$, with $A := [T]_\mathcal{B}$ upper triangular.
We have

$$Tv_1 = A_{11}v_1 + A_{21}v_2 + \cdots + A_{n1}v_n = A_{11}v_1$$

since A is upper triangular. Therefore, we see v_1 is an eigenvector of T, with eigenvalue A_{11}. Therefore, by the previous proposition, we have $T^*v_1 = \overline{A_{11}}v_1$ (using the fact that T is normal). But $[T^*]_\mathcal{B} = [T]_\mathcal{B}^* =$

A^*, so $T^*v_1 = \sum_{j=1}^n A^*_{j1}v_j = \sum_{j=1}^n \overline{A_{1j}}v_j$. But we know that $\overline{A_{11}}v_1 = \overline{A_{11}}v_1 + \overline{A_{12}}v_2 + \cdots + \overline{A_{1n}}v_n$, but by linear independence of $(v_1, \dots, v_n)$, this means that $\overline{A_{1j}} = A_{1j} = 0$ for all $j > 1$.
Therefore, the top row of A away from the diagonal entry is zero. Iterating the argument, $Tv_2 = \sum_{j=1}^n A_{j2}v_2 = A_{22}v_2$ (since $A_{12} = 0$), but $T^*v_2 = \overline{A_22}v_2$ and $\overline{A_{22}}v_2 = \sum_{j=1}^n \overline{A_{2j}}v_j$, which by linear independence of $(v_1, \dots, v_n)$ means that $\overline{A_{2j}} = A_{2j} = 0$ for all $j > 2$. Repeating the argument for all rows, we get that A is a diagonal matrix. ■

Proposition:
Let $F = \mathbb{C}$. Then V has an orthonormal basis $\mathcal{B}$ such that $[T]_\mathcal{B}$ is upper triangular.
Proof:
By induction on $\dim V$. The base case of $\dim V = 1$ is clear.
Now, inductively suppose $\dim V > 1$ and let λ be an eigenvalue of T^*, which exists since $F = \mathbb{C}$. Thus, there exists $v \neq 0$ such that $T^*v = \lambda v$ for some $v \neq 0$. Let $W = \mathrm{Span}(v)$. W is T^*-invariant as v is an eigenvector of T^*. Therefore, $W^\perp$ is T-invariant. Use this to do the (not necessarily T-invariant) direct sum decomposition $V = W^\perp \oplus W$.
But now consider $T|_{W^\perp}$, the linear operator (by invariance) on the $\dim V - 1$ space $W^\perp$. Therefore, by the induction hypothesis, there is an orthonormal basis $\mathcal{B}' = (v_1, \dots, v_{n-1})$ for $W^\perp$ such that $[T|_{W^\perp}]_{\mathcal{B}'}$ is upper triangular. Let $\mathcal{B} = (v_1, \dots, v_{n-1}, \frac{v}{||v||})$. Since $v_1, \dots, v_{n-1} \in W^\perp$, this basis is orthogonal, and orthonormal by construction. Therefore, $\mathcal{B}$ is an orthonormal basis for $\mathcal{B}$.
Therefore, by the direct sum decomposition, we have $[T]_\mathcal{B} = \begin{bmatrix} (T|_{W^\perp})_{\mathcal{B}'} & * \\ 0 & * \end{bmatrix}$, where we don't know the last column since the decomposition is not T-invariant (so it's not block diagonal). But now we see this is upper triangular, since $[T|_{W^\perp}]_{\mathcal{B}'}$ is upper triangular, and the last column does not matter. ■

Corollary:
If $F = \mathbb{C}$, and T is normal then V has an orthonormal basis of eigenvectors of T.
Proof:
Apply the previous two propositions together. ■

Corollary:
If $F = \mathbb{C}$ then T is normal if and only if V has a basis of eigenvectors for T.
Proof:
The forward direction is the previous corollary, the reverse direction we proved initially. ■

Summary:

$$\begin{array}{l} \quad\text{T is self-adjoint} \\ (1)\downarrow \quad \uparrow (3) \text{ if } F = \mathbb{R} \\ \quad\text{T is orthonormally diagonalizable} \\ (2)\downarrow \quad \uparrow (4) \text{ if } F = \mathbb{C} \\ \quad\text{T is normal} \end{array}$$

Example:
(3) above is false in general if $F = \mathbb{C}$. Let $T : \mathbb{C} \to \mathbb{C}$ be the linear operator corresponding to multiplication by i. Note that $1 \in \mathbb{C}$ is an eigenvector with eigenvalue i. But $T \neq T^*$ since T^* is multiplication by $-i$.

Exercise:
(3) holds if all eigenvalues are real.

Example:
(4) is false in general if $F = \mathbb{R}$. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by $-\frac{\pi}{2}$. T is unitary since it preserves norm, hence T is normal. But T has no eigenvectors, so not orthonormally diagonalizable.

Let's now look at the matrix versions of these theorems:

Theorem:
(1) says if $A \in M_n(F)$ is Hermitian, then A is unitarily diagonalizable.
(4) says if $A \in M_n(\mathbb{C})$ is normal then it is unitarily diagonalizable.
Proof:
Let $T : F^n \to F^n$ be the linear operator such that $[T] = A$. Fix the standard inner product on F^n. Then:
(1) Suppose A is Hermitian, so $A = A^*$. Therefore, $[T] = A$, and $[T*] = [T]^* = A^*$ since the standard basis is orthonormal. Therefore, $T = T^*$ since they agree as matrices on a basis. Therefore, T is self-adjoint, so there is an orthonormal basis $\mathcal{B}$ for F^n such that $\mathcal{B}$ is made of eigenvectors of T, i.e. $D := [T]_{\mathcal{B}}$ is diagonal. So D and A represent the same linear operator with respect to (possibly) different orthonormal bases. Therefore, by a previous theorem, we have that $A = P^{-1}DP$ where P is unitary, so A is unitarily diagonalizable.
(4) is similar (exercise). $\blacksquare$

We can rephrase our orthonormally diagonalizable results intrinsically in terms of projections.

Theorem (Spectral Theorem):
Let $(V, \langle , \rangle)$ be a finite dimensional inner product space and $T : V \to V$ a linear operator. Suppose T is either self-adjoint or $F = \mathbb{C}$ and T is normal. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T, and set $W_1, \ldots, W_k$ to be the corresponding eigenspaces. Then,
(1) $V = W_1 \oplus \cdots \oplus W_k$
(2) For $i \neq j$, W_j is orthogonal to W_i, meaning that every vector in W_i is orthogonal to every vector in W_j.
(3) Letting E_i be the projections onto W_i, we have $T = \lambda_1 E_1 + \cdots + \lambda_k E_k$.
Proof:
We know from our main theorems that T has an orthonormal basis of eigenvectors. In particular, T is diagonalizable and so (a) and (c) follow. It suffices to prove (b).
Fix $i \neq j$, $v \in W_i$, $w \in W_j$. Then,

$$\begin{aligned}
\lambda_i \langle v, w\rangle &= \langle \lambda_i v, w\rangle \\
&= \langle Tv, w\rangle \\
&= \langle v, T^* w\rangle \\
&= \begin{cases} \langle v, Tw\rangle, & T \text{ self-adjoint} \\ \langle v, T^* w\rangle & T \text{ normal} \end{cases} \\
&= \begin{cases} \langle v, \lambda_j w\rangle, & T \text{ self-adjoint} \\ \langle v, \overline{\lambda_j} w\rangle & T \text{ normal, by a theorem using normality and } F = \mathbb{C} \end{cases} \\
&= \begin{cases} \overline{\lambda_j}\langle v, w\rangle, & T \text{ self-adjoint} \\ \lambda_j \langle v, w\rangle, & T \text{ normal} \end{cases} \\
&= \lambda_j \langle v, w\rangle \quad \text{Since eigenvalues of self-adjoint operator are real} \\
&\implies (\lambda_i - \lambda_j)\langle v, w\rangle = 0 \\
&\implies \langle v, w\rangle = 0 \quad \blacksquare
\end{aligned}$$

Remark:
The converse is also true: if T satisfies (a) and (b) then T is normal and if $F = \mathbb{R}$ then T is self-adjoint.

Proof:
By (a) we have a basis of eigenvectors, we can normalize to make it normal, and (b) says they are orthonormal. Therefore T is normal and self-adjoint if $F = \mathbb{R}$. $\blacksquare$

Forms

Definition:
Let V be a vector space, with $F = \mathbb{R}$ or $F = \mathbb{C}$. A form on V is a function $f : V \times V \to F$ satisfying:
(1) $f(\lambda v + w, u) = \lambda f(v, u) + f(w, u)$ for all $v, w, u \in V$, $\lambda \in F$ (it is linear in the first argument, or in other words, $f(*, u)$ is a linear functional on V)
(2) $f(v, \lambda w + u) = \overline{\lambda} f(v, w) + f(v, u)$ for all $v, w, u \in V$, $\lambda \in F$ (it is conjugate linear in the second argument).
If $F = \mathbb{R}$, we call f a bilinear form, and if $F = \mathbb{C}$, we call f a sesquilinear form.

Example:
Any inner product on V is a form. Note that a form need not be an inner product, since we do not require conjugate symmetry and hence it need <u>not</u> be the case that $f(v, v) \in \mathbb{R}$, let alone that $f(v, v)$ is positive for $v \neq 0$.

Definition:
A form is Hermitian if $f(v, w) = \overline{f(w, v)}$ for all $v, w \in V$ (also called symmetric if $F = \mathbb{R}$). So inner products are Hermitian forms with a positivity condition.

Example:
Suppose $\langle \cdot, \cdot \rangle$ is an inner product on V and $T : V \to V$ is a linear operator. Then $f(v, w) = \langle Tv, w \rangle$ is a form. Note if $v \neq 0 \in \text{null}(T)$, then $f(v, v) = \langle Tv, v \rangle = 0$ so f is not an inner product in this case.
Proof:

$$f(\lambda v + u, w) = \langle T(\lambda v + u), w \rangle = \langle \lambda Tv + Tu, w \rangle = \lambda \langle Tv, w \rangle + \langle Tv, w \rangle = \lambda f(v, w) + f(u, w)$$

Axiom (2) of f holds as it holds for $\langle \cdot, \cdot \rangle$. $\blacksquare$

Theorem:
Suppose V is a finite dimensional vector space and $\langle \cdot, \cdot \rangle$ is an inner product on V. Then, if f is a form on V, then there is a unique linear operator T_f such that $f(v, w) = \langle T_f v, w \rangle$ for all $v, w \in V$.
Proof:
To prove uniqueness, suppose $f(v, w) = \langle Tv, w \rangle$ for some linear operator T. We will show T is uniquely determined by this identity. Fix an orthonormal basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V. Then, for all $v \in V$, $v = \langle v, v_1 \rangle v_1 + \cdots + \langle v, v_n \rangle v_n$. In particular for any $j = 1, \ldots, n$, $Tv_j = \langle Tv_j, v_1 \rangle v_1 + \cdots + \langle Tv_j, v_n \rangle v_n$, implying the (i, j) entry of $[T]_{\mathcal{B}}$ is $\langle Tv_j, v_i \rangle = f(v_j, v_i)$, so $[T]_{\mathcal{B}}$ and hence T is determined by $f(v_j, v_i)$ for $i, j = 1, \ldots, n$, proving uniqueness.
To prove existence, fix $w \in V$ and consider the functional $f(\cdot, w) : V \to F$. Then, there is a unique vector $w' \in V$ such that $f(*, w) = \langle *, w' \rangle$ (by Riesz). Let $S : V \to V$ be the map determined by $w \mapsto w'$, i.e. $f(*, w) = \langle *, S(w) \rangle$, which is well-defined by uniqueness. As an exercise, check linearity. Then, let $T = S^*$. Then, for any $v, w \in V$,

$$f(v, w) = \langle v, S(w) \rangle = \langle v, T^* w \rangle = \langle Tv, w \rangle$$

which is what we wanted. $\blacksquare$

Continuing with the discussion of sesquilinear forms f on finite dimensional vector spaces, let $\text{Form}(V)$ be the set of all forms on V. This is an F-vector space: $(f + g)(v, w) = f(v, w) + g(v, w)$, $(\lambda f)(v, w) = \lambda f(v, w)$.

Let $\mathrm{End}(V)$ be the F-vector space of all linear operators on V.

Proposition:
Fix an inner product $\langle \cdot, \cdot \rangle$ on V. Then, the map $f \mapsto T_f$ from $\mathrm{Form}(V)$ to $\mathrm{End}(V)$ is an isomorphism of vector spaces.
Proof:
As an exercise, check linearity: $T_{f+g} = T_f + T_g$ and $T_{\lambda f} = \lambda T_f$.
For surjectivity, let $T \in \mathrm{End}(V)$. Define $f(v,w) := \langle Tv, w \rangle$ so $f \in \mathrm{Form}(V)$, so by uniqueness in the previous proposition, $T = T_F$.
For injectivity, suppose $T_f = T_g$. Then, this means

$$\langle T_f v, w \rangle = \langle T_g v, w \rangle \implies f(v,w) = g(v,w) \implies f = g$$

for all $v, w \in V$. ■

Remark:
If $f = \langle \cdot, \cdot \rangle$, then $T_f = I$.

Proposition:
Let $(V, \langle, \rangle)$ be a finite dimensional inner product space and let $f \in \mathrm{Form}(V)$. Then, f is Hermitian if and only if T_f is self-adjoint.
Proof:
Let $v, w \in V$. Then,

$$\begin{aligned} f \text{ hermitian} &\iff f(v,w) = \overline{f(w,v)} \\ &\iff \langle T_f v, w \rangle = \overline{\langle T_f w, v \rangle} \\ &\iff \langle v, T_f^* w \rangle = \langle v, T_f w \rangle \quad (1) \end{aligned}$$

But now fix $w \in V$. Let $(v_1, \ldots, v_n)$ be an orthonormal basis for $(V, \langle, \rangle)$. Then,

$$T_f w = \langle T_f w, v_1 \rangle v_1 + \cdots + \langle T_f w, v_n \rangle v_n$$

and

$$T_f^* w = \langle T_f^* w, v_1 \rangle + \cdots + \langle T_f^* w, v_n \rangle v_n$$

Apply (1) (to $(v_1, \ldots, v_n)$) and the unique representation of the basis elements with the above equations to get that we have

$$(1) \iff T_f w = T_f^* w \iff T_f = T_f^*$$

since w was arbitrary. ■

Let's try to use the isomorphism of forms with operators to add some concreteness. An example:

Proposition:
Let $F = \mathbb{C}$. Suppose $f \in \mathrm{Form}(V)$. Then f is Hermitian if and only if $f(v,v) \in \mathbb{R}$ for all $v \in V$.
Proof:
In the forward direction, f Hermitian implies $f(v,v) = \overline{f(v,v)}$, so we must have $f(v,v) \in \mathbb{R}$ by definition of conjugate.
In the reverse direction, we have

$$f(v+w, v+w) = f(v,v,) + f(v,w) + f(w,v) + f(w,w)$$

Since $f(v+w, v+w)$, $f(v,v)$, $f(w,w) \in \mathbb{R}$, by closure of the field operations, we must have $f(v,w) + f(w,v) \in \mathbb{R}$.
Doing similarly with $v + iw$,

$$f(v+iw, v+iw) = f(v,v) - if(v,w) + if(v,w) + f(w,w)$$

and with an identical argument, since $f(v+iw, v+iw)$, $f(v,v)$, $f(w,w) \in \mathbb{R}$, we must have $-if(v,w) + if(v,w) \in \mathbb{R}$.
Combining these two, since $f(v,w) + f(w,v) \in \mathbb{R}$, we must have

$$f(v,w) + f(w,v) = \overline{f(v,w)} + \overline{f(w,v)}$$

(since the conjugate is additive), and we get that

$$-if(v,w) + if(w,v) = i\overline{f(v,w)} - i\overline{f(w,v)}$$

Add i times the second equation with the first, and get that $2f(v,w) = 2\overline{f(w,v)}$, so $f(v,w) = \overline{f(w,v)}$ so f is Hermitian. ■

Corollary:
Suppose $F = \mathbb{C}$, with V a finite dimensional inner product space. Let $T \in \mathrm{End}(V)$. Then T is self-adjoint if and only if $\langle Tv, v\rangle \in \mathbb{R}$ for all $v \in V$.
Proof:
Let $f \in \mathrm{Form}(V)$ such that $T = T_f$. Then,

$$T = T^* \iff f \text{ Hermitian} \iff f(v,v) \in \mathbb{R} \iff \langle Tv, v\rangle \in \mathbb{R}$$

for all $v \in V$. ■

Definition:
Let $f \in \mathrm{Form}(V)$, $\mathcal{B} = (v_1, \dots, v_n)$ a basis for V. The matrix of f with respect to $\mathcal{B}$, denoted $[f]_{\mathcal{B}}$, is the matrix whose (i,j) entry is $f(v_j, v_i)$.

Exercise:
If we have an inner product on V and $\mathcal{B}$ is orthonormal, then $[f]_{\mathcal{B}} = [T_f]_{\mathcal{B}}$.

Theorem (Principal Axis Theorem):
Let $(V, \langle\cdot,\cdot\rangle)$ be a finite dimensional inner product space. Let $f \in \mathrm{Form}(V)$. If f is Hermitian, then there exists an orthonormal basis $\mathcal{B}$ with respect to which $[f]_{\mathcal{B}}$ is diagonal with real entries.
Proof:
Let $T = T_f$. Since f is Hermitian, T is self-adjoint. By the spectral theorem, there exists an orthonormal basis $\mathcal{B}$ for V of eigenvectors for T. Therefore, $[T]_{\mathcal{B}}$ is diagonal, therefore by the previous exercise, $[f]_{\mathcal{B}}$ is diagonal. But $[T]_{\mathcal{B}} = [f]_{\mathcal{B}}$ has real eigenvalues, thus real diagonal entries, since T is self-adjoint. ■

Let V be a finite dimensional vector space, where $f \in \mathrm{Form}(V)$. How can we tell if f is an inner product? Fix an arbitrary basis $\mathcal{B} = (v_1, \dots, v_n)$ of V, and consider the matrix of f with respect to $\mathcal{B}$, $A := [f]_{\mathcal{B}}$, $A_{ij} = f(v_j, v_i)$.

Definition:
A form is positive if it is an inner product, i.e. f is Hermitian and $f(v,v) > 0$ for all $v \neq 0$.

Definition:
Given $A \in M_n(F)$, we can define a form on F^n by $g_A(x,y) = y^*Ax$ for all $x, y \in F^n$. It's easy to check that this gives a sesquilinear form.

Remark:
Let $f \in \mathrm{Form}(V)$, $\mathcal{B}$ be a basis for V, $A := [f]_{\mathcal{B}}$, and $n := \dim V$. What is the relationship between f on V and g_A on F^n? Let $\mathcal{B} = (v_1, \dots, v_n)$. Given $v, w \in V$, $v = a_1v_1 + \cdots + a_nv_n$, and $w = b_1v_1 + \cdots + b_nv_n$, then $f(v,w) = g_A((a_1, \dots, a_n)^T, (b_1, \dots, b_n)^T)$.

Proof:

$$f(v,w) = f(\sum_i a_i v_i, \sum_i b_i v_i) = \sum_{i,j} a_i \overline{b_j} f(v_i, v_j) = \sum_{i,j} a_i \overline{b_j} A_{ji} = (\overline{b_1}, \dots, \overline{b_n}) A (a_1, \dots, a_n)^T \quad \blacksquare$$

Corollary:
f is positive on V if and only if g_A is positive on F^n. $\blacksquare$

Proposition:
g_A is positive if and only if $A = P^*P$ for some invertible matrix $P \in M_n(F)$.
Proof:
Suppose $A = P^*P$ for $P \in M_n(F)$ invertible. Then

$$\overline{g_A(x,y)} = \overline{y^*Ax} = \overline{y^*P^*Px} = y^T P^T \overline{P}\overline{x} = \overline{x}^T (P^T\overline{P})^T y = x^*(P^*P)y = x^*Ay = g_A(y,x)$$

for all $x, y \in F^n$. Therefore g_A is Hermitian. Fix $x \neq 0 \in F^n$. Then,

$$g_A(x,x) = x^*Ax = x^*P^*Px = (Px)^*(Px) = \langle Px, Px \rangle$$

where $\langle \cdot, \cdot \rangle$ is the standard inner product. But P is invertible, so $Px \neq 0$ since $x \neq 0$, so by positivity of the inner product, we have that $\langle Px, Px \rangle > 0$, showing positive-definiteness. Thus, g_A is positive.
Conversely, suppose g_A is positive, therefore g_A is an inner product on F^n. Let $\mathcal{B} = (q_1, \dots, q_n)$ be an orthonormal basis of F^n with respect to g_A. Then, $g_A(q_i, q_j) = q_j^* A q_i = \delta_{ij}$. Form the matrix Q whose columns are $q_1, \dots, q_n$. Then, $Q^*AQ = I$ by the prior equation. Since the columns of Q form a basis of F^n, Q is invertible. Thus, we can multiply the above equation by ${Q^*}^{-1}$ on the left and Q^{-1} on the right to get that $A = ({Q^*}^{-1})Q^{-1} = (Q^{-1})^*Q^{-1}$, so we see $P = Q^{-1}$ works. $\blacksquare$

Corollary:
Let $F \in \mathrm{Form}(V)$. f is positive if and only if for some (equivalently any) basis $\mathcal{B}$ of V, $[f]_{\mathcal{B}} = P^*P$ for some invertible $P \in M_n(F)$. $\blacksquare$

Corollary:
Let $F \in \mathrm{Form}(V)$. If f is positive and $\mathcal{B}$ is a basis for V, then $\det([f]_{\mathcal{B}}) > 0$.
Proof:

$$\det([f]_{\mathcal{B}}) = \det(P^*P) = \det(P^*)\det(P) = \overline{\det(P)}\det(P) = |\det(P)|^2 > 0$$

since P is invertible, so its determinant is nonzero. $\blacksquare$

This is not sufficient: Take $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. This has positive determinant but is the negative of the matrix of the standard inner product and thus cannot be positive.

We now try to generalize this condition to get a sufficient one.

Definition:
Let $A \in M_n(F)$. The principal minors of A are the determinants of the $k \times k$ matrices for $1 \leq k \leq n$, given by

$$\Delta_k(A) = \begin{vmatrix} A_{11} & A_{12} & \dots & A_{1k} \\ \vdots & & & \vdots \\ A_{k1} & A_{k2} & \dots & A_{kk} \end{vmatrix}$$

Proposition:
Let $f \in \mathrm{Form}(V)$, where $\mathcal{B} = (v_1, \dots, v_n)$ is a basis of V. If f is positive, then $\Delta_k([f]_{\mathcal{B}}) > 0$ for all

$k = 1, \dots, n$.
Proof:
Let $\mathcal{B}_k = (v_1, \dots, v_k)$. Then, $V_k := \mathrm{Span}(\mathcal{B}_k)$ is a k dimensional subspace of V with basis $\mathcal{B}_k$. Since f is a positive form on V, $f_k := f|_{V_k \times V_k} : V_k \times V_k \to F$ is also a positive form (this time on V_k). Hence, by the previous corollary, $\det([f_k]_{\mathcal{B}_k}) > 0$. But $[f_k]_{\mathcal{B}_k}$ is the same matrix as the one we take the determinant of to get the minor, giving the result. ■

The converse turns out to also be true.

Lemma:
Let $A \in M_n(F)$ be invertible. The following are equivalent:
(1) There exists an upper triangular matrix $P \in M_n(F)$ with 1's on the diagonal such that AP is lower triangular.
(2) $\Delta_k(A) \neq 0$ for all $k = 1, \dots, n$.
Proof:
(2) implies (1): We want to find $P \in M_n(F)$ with the given properties. We will write $P = (P_{ij})$ and view P_{ij} as indeterminates subject to the constraints imposed by (2). We have $P_{ij} > 0$ if $i > j$, and $P_{ii} = 1$ for all i. We need conditions on P_{ij} for $j > i$. Let $B = AP$. Then $B_{jk} = \sum_{r=1}^{n} A_{jr}P_{rk}$, implying $B_{jk} = \sum_{j=1}^{k-1} A_{jr}P_{rk} + A_{kk}$ by the prior conditions. So B is lower triangular if and only if $B_{jk} = 0$ for $j < k$ if and only if $\sum_{r=1}^{k-1} A_{jr}P_{rk} = -A_{kk}$ for $i \leq j \leq k-1$. For fixed k, this condition is a system of $k-1$ linear equations in the $k-1$ unknowns $P_{1k}, \dots, P_{k-1,k}$ with coefficient matrix the matrix of the minor $\Delta_{k-1}(A)$. Thus, the system has a unique solution for each k since its determinant, the minor Δ_{k-1}, is nonzero (the matrix representing the system is invertible, so it has a unique solution). Therefore, such a P with the desired properties exists (uniquely).
(1) implies (2): Let $B = AP$. Let $A_1, \dots, A_n$ be the columns of A, and $B_1, \dots, B_n$ be the columns of B. Because P is upper triangular with 1s on the diagonal, looking at the matrix multiplication, we have $B_1 = A_1$, $B_2 = P_{12}A_1 + A_2$, and in general, $B_n = P_{1n}A_1 + \cdots + P_{n-1,n}A_{n-1} + A_n$. These are just column operations on A, so we see we get B from column operations on A, therefore we have $\det B = \det A$. Thus, $\Delta_n(B) = \Delta_n(A)$.
But this argument works if we truncate our above matrix multiplication to size $k \times k$ (i.e. cut off the columns of both matrices), since P is upper triangular so its lower entries don't matter. Therefore, we get that $\Delta_k(B) = \Delta_k(A)$ for all $k = 1, \dots, n$. But B is lower triangular, so each of its principal minors is $\Delta_k(B) = B_{11} \dots B_{kk}$ (the product of the diagonal entries, since each minor matrix is still lower triangular). But this product cannot vanish, since B itself is lower triangular and invertible (it is the product of P and A that are invertible), so its determinant is the product of its diagonal entries, and thus none can be zero since B is invertible. Thus, $\Delta_k(A) = \Delta_k(B) \neq 0$. ■

Let's extract the above argument:

Lemma:
If $A \in M_n(F)$ and $P \in M_n(F)$ is upper triangular with 1s on the diagonal, then $\Delta_k(AP) = \Delta_k(A)$ for all $k = 1, \dots, n$. Similarly, if $Q \in M_n(F)$ is lower triangular with 1s on the diagonal, then $\Delta_k(QA) = \Delta_k(A)$ for $k = 1, \dots, n$ (just work with rows instead of columns in the proof). ■

Lemma:
Let $f \in \mathrm{Form}(V)$. Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{B}' = (w_1, \dots, w_n)$ be bases of V. Then $[f]_{\mathcal{B}'} = P^*[f]_{\mathcal{B}}P$, where $P = (P_{ij})$ is such that $w_j = \sum_{i=1}^{n} P_{ij}v_i$.
Proof:
Compute

$$([f]_{\mathcal{B}'})_{jk} = f(w_k, w_j) = f(\sum_{s=1}^{n} P_{sk}v_s, \sum_{r=1} P_{rj}v_r) = \sum_{r,s} P_{sk}\overline{P_{rj}}f(v_s, v_r)$$

$$= \sum_{r,s} P_{sk}\overline{P_{rj}}([f]_{\mathcal{B}})_{rs} = \sum_{r,s} \overline{P_{rj}}([f]_{\mathcal{B}})_{rs}P_{sk} = (P^*[f]_{\mathcal{B}}P)_{jk} \quad \blacksquare$$

Theorem:
Let $f \in \text{Form}(V)$, and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V, and $A = [f]_{\mathcal{B}}$. Then f is positive if and only if $A^* = A$ and $\Delta_k(A) > 0$ for all $k = 1, \dots, n$.
Proof:
The forward direction was done previously. In the reverse direction, $A^* = A$ tells us f is Hermitian, thus we only need to prove the condition on its minors gives us positive-definiteness, i.e. $f(v, v) > 0$ for all $v \neq 0$. Since $\Delta_k(A) \neq 0$ (they are greater than 0) for all k, there is an upper triangular $P \in M_n(F)$ with 1s on the diagonal such that AP is lower triangular by the lemma. But we know P^* is lower triangular since P is upper triangular, therefore P^*AP is lower triangular (as a product of lower triangular matrices). But

$$(P^*AP)^* = P^*A^*P = P^*AP$$

since $A^* = A$, so P^*AP is Hermitian. Therefore, we must have that $D := P^*AP$ is diagonal (as that is the only case when a lower triangular matrix can be "symmetric") with real diagonal entries (since it is Hermitian, and those diagonal entries are its eigenvalues). But now by the previous lemmata, for each $k = 1, \dots, n$,

$$\Delta_k(D) = \Delta_k(P^*AP) = \Delta_k(AP) = \Delta_k(A) > 0$$

Therefore, we see that each diagonal entry $D_{ii} > 0$ (looking at each individual minor one by one, as the product of diagonal entries). But D and A represent the same form with different bases by the previous lemma. Let $\mathcal{B}' = (w_1, \dots, w_n)$ be the basis that D represents f in, and let $v \neq 0 \in V$. Write $v = a_1w_1 + \cdots + a_nw_n$, so

$$f(v, v) = (\overline{a_1}, \dots, \overline{a_n})D(a_1, \dots, a_n)^T = (\overline{a_1}, \dots, \overline{a_n})(D_{11}a_1, \dots, D_{11}a_n)^T = |a_1|^2D_{11} + \cdots + |a_n|^2D_{nn} > 0$$

since $D_{ii} > 0$ and not all a_i's are zero since $v \neq 0$. Thus we see f is positive definite. $\blacksquare$